

Introduction for students: why Chapter 4 matters

Chapter 4 turns circuit analysis into a method. Instead of choosing equations by hand, you will write exactly the equations a circuit needs, in a fixed order, and solve them. Chapter 4 runs from week 5 to week 9, the largest share of the material on the final exam. Part 1 (weeks 5–6) covers the two methods every later topic relies on: node voltages and mesh currents.

What you will learn in Part 1

- **Counting:** essential nodes, essential branches and meshes tell you how many equations a circuit needs, before you write any.
- **The node-voltage method:** pick a reference node, then write KCL at every other essential node, with each current written from the node voltages.
- **Dependent sources:** add one constraint equation that expresses the controlling voltage or current through your unknowns.
- **Special cases with voltage sources:** a source to the reference fixes a node voltage; a source between two non-reference essential nodes makes a supernode.
- **The mesh-current method:** give each mesh a clockwise current and write KVL around it; a shared resistor carries the difference of two mesh currents.
- **Special cases with current sources:** a current source in one mesh fixes that mesh current; a source shared by two meshes makes a supermesh.
- **Choosing a method:** count the equations each method needs and pick the one with fewer.

Later in this course

Topic	Where Part 1 comes back
Part 2: source transformations and superposition	Both are checked by solving the same circuit with node voltages or mesh currents
Part 2: Thévenin and Norton equivalents	The open-circuit voltage, the short-circuit current and the test-source method are all found with node or mesh equations
Chapter 5: Operational amplifiers	Op-amp circuits are solved with node equations written at the op-amp inputs
Chapter 7: First-order circuits	The resistance seen by a capacitor or an inductor is found with the same methods

In your program

- **Electrical and Electronics:** power-flow studies of a national grid write node equations at every bus (a node of the network, such as a substation), the node-voltage method on a very large scale.
- **Computer, Communication and Telecom:** circuit simulators used to design chips, such as SPICE (developed at UC Berkeley and first presented in 1973), solve circuits by writing node equations, an extended form of this week's method.
- **Biomedical:** models of tissue and electrode interfaces are networks of resistors and sources solved node by node.
- **Mechanical and Industrial:** heat flows through walls and pipes are modelled as resistor networks, and the same node equations give the temperatures.

- **Surveying:** instrument circuits with several batteries and loads are checked with the same node and mesh equations.

As an engineer

- **Count before you write.** Two minutes spent counting essential nodes and meshes, and spotting supernodes and supermeshes, often halves the algebra.
- **Signs are a convention, not a guess.** Currents leaving a node are positive; voltage drops around a mesh are positive. Keep the rule and the signs take care of themselves.
- **Check without solving again.** Substitute your answers into KCL at each node, reading every direction from the figure. A power balance catches most slips too, but not a polarity misread the same way throughout.

In real life

- **Jump-starting a car.** A good battery, a flat battery and the starter motor all connect to the same two terminals: one node equation tells you whether the flat battery is being charged or is helping to crank the engine (S15).
- **Every simulator you will use.** Circuit simulation software writes the node equations of Part 1 automatically for circuits with thousands of nodes. Knowing the method lets you judge whether its answer makes sense.

How to study this part, and where AI fits

- AI tools are allowed on the practice sheet. Use them to check your work, not to replace it.
- In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many (arXiv 2512.10159)¹. In node and mesh equations these slips show up as a current entered with the wrong sign, a shared resistor written with the mesh currents the wrong way round, or a current direction misread when finding a power. S11 and S12 show such slips: find them before you trust any answer.
- Check every answer yourself: KCL at each node and a power balance need no device.
- The quiz is device-free, and every question is a twin of a practice-sheet question. If you can do the sheet on your own, you will do well on the quiz.

Checklist: you can do these without help

- ☐ Count the essential nodes and meshes, and say how many equations each method needs
- ☐ Write node-voltage equations, including a dependent source, a known node voltage and a supernode
- ☐ Write mesh-current equations, including a dependent source, a known mesh current and a supermesh
- ☐ Choose the method with fewer equations, and say why
- ☐ Find the power of every source and say whether it absorbs or delivers
- ☐ Check a solution with KCL at each node, KVL around an unused loop, or a power balance

¹<https://arxiv.org/html/2512.10159>